UNIT - II

Ultrasonic Testing (UT)

1. Introduction

* Ultrasonic testing uses high frequency sound waves (0.5 - 25 MHz) to detect internal and surface flaws, measure thickness and find material properties.
* Sound above the human hearing limit (above 20 kHz) is called ultrasound .
* A short pulse of sound is sent into the part; the reflected (echo) signal tells the position and size of the flaw.
* Needs access to only one side of the part (pulse-echo) - big advantage over radiography.

Frequency Ranges of Sound

Type	Frequency	Remark
Infrasonic	below 20 Hz	not audible
Audible	20 Hz - 20 kHz	human ear
Ultrasonic	above 20 kHz	used in NDT (0.5 - 25 MHz)

Basic Principle

* Sound waves travel through a homogeneous medium at constant velocity and get reflected at an interface (crack, void, back wall) where the acoustic property changes.
* Time of flight of echo gives the distance: $d = v\,t / 2$ (v = velocity of sound, t = total round trip time).
* Amplitude of echo gives an idea of flaw size .

Pulser sends electrical pulse → Probe converts to ultrasound → Sound travels in test piece → Echo from flaw / back wall

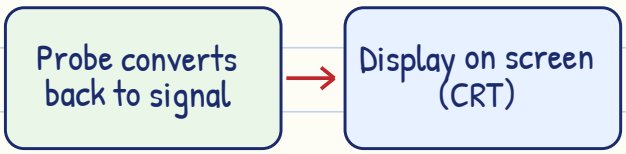


Fig: Working of pulse-echo ultrasonic testing

2. Principle of Wave Propagation

* A wave is a **disturbance** that transfers energy through a medium without permanent movement of particles.
* **Wavelength (lambda):** distance between two successive crests.
* **Frequency (f):** number of cycles per second, in Hz.
* **Velocity (v):** $v = f \times lambda$; depends on the material, not on frequency.
* **Time period** $T = 1/f$; **Amplitude:** maximum particle displacement.

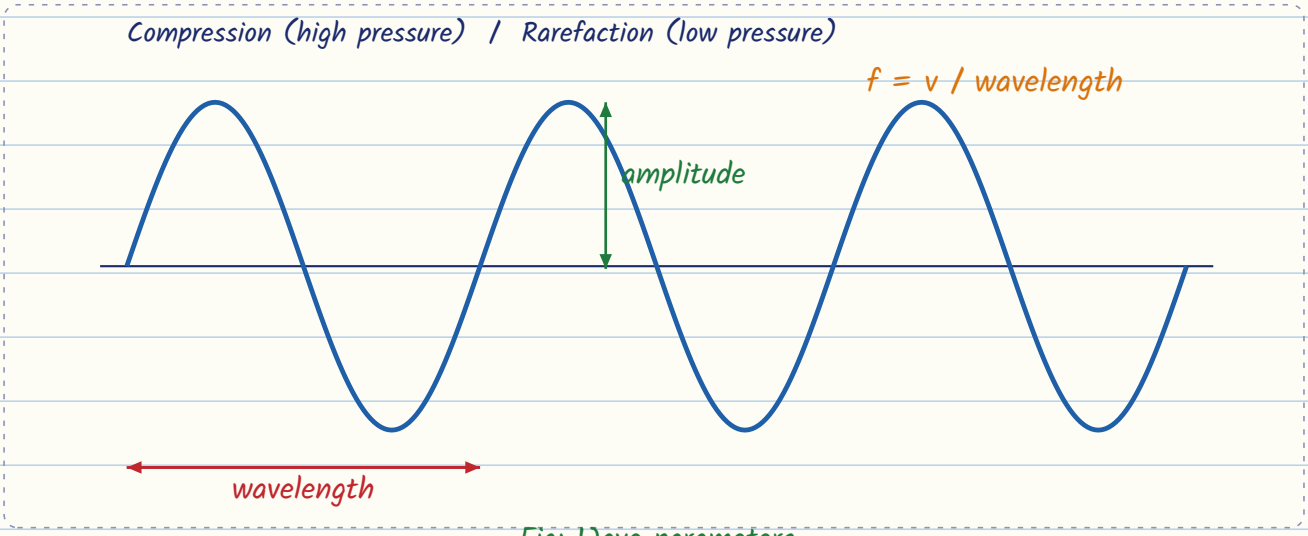


Fig: Wave parameters

Modes (Types) of Ultrasonic Waves

* **Longitudinal (compression) wave:** particle motion is **parallel** to direction of wave travel. Travels in solids, liquids and gases. Fastest wave. Used in straight-beam testing.
* **Transverse (shear) wave:** particle motion is **perpendicular** to wave direction. Only in solids. Velocity is about **half** of longitudinal velocity.
* **Surface (Rayleigh) wave:** elliptical particle motion, travels along the surface to a depth of about **one wavelength**. Detects surface and near-surface cracks.
* **Plate (Lamb) wave:** in thin plates (thickness less than a few wavelengths); symmetric and asymmetric modes.

Wave	Particle motion	Medium	Velocity
Longitudinal	Parallel to propagation	Solid, liquid, gas	Highest (steel: 5900 m/s)
Shear	Perpendicular	Solid only	About 0.5 x longitudinal (3200 m/s)
Surface (Rayleigh)	Elliptical	Surface of solids	About 0.9 x shear (2950 m/s)
Lamb	Complex	Thin plates	Depends on thickness, frequency

Velocity of Sound in Materials

Material	Longitudinal velocity (m/s)
Steel	5900
Aluminium	6300
Copper	4700
Water	1480
Air	330
Perspex	2700

* **Longitudinal velocity formula:** v = sqrt(E (1 - nu) / [rho (1 + nu)(1 - 2 nu)]) where E = Young's modulus, rho = density, nu = Poisson's ratio.
* **Shear velocity:** v = sqrt(G / rho) where G = shear modulus.

3. Acoustic Impedance, Reflection and Transmission

* **Acoustic impedance** Z = rho x v (density x velocity). Unit: kg / m^2 s (Rayl).
* When a wave meets an interface of two media with different impedance, part is **reflected** and part is **transmitted**.
* **Reflection coefficient:** R = (Z2 - Z1)^2 / (Z2 + Z1)^2 ; Transmission coefficient T = 1 - R = 4 Z1 Z2 / (Z1 + Z2)^2.
* Greater the difference in Z -> larger the reflection.
* **Steel / air interface:** almost 100% reflection (R about 0.99998) - so air gap between probe and part must be removed with a **couplant**.

* **Steel / water:** about 94% transmitted - water is a good coupling medium.
* A crack or void filled with air gives a **very strong echo** .

Remember:

Why couplant? Air gap between probe and rough surface reflects almost the entire energy. Couplants: water, oil, grease, glycerine, gel, cellulose paste.

4. Reflection, Refraction, Diffraction

4.1 Reflection

* At normal incidence the wave goes back along the same path. **Angle of incidence = angle of reflection** for oblique incidence.
* Used in pulse-echo method. Smooth flat interface -> specular reflection; rough surface scatters the sound.

4.2 Refraction - Snell's Law

* Sound passing obliquely from one medium to another **bends** due to the change in velocity.
* **Snell's law:** sin(theta1) / sin(theta2) = v1 / v2.
* Used to produce **angle beam (shear wave) probes** - Perspex wedge on steel at 45, 60, 70 degrees.

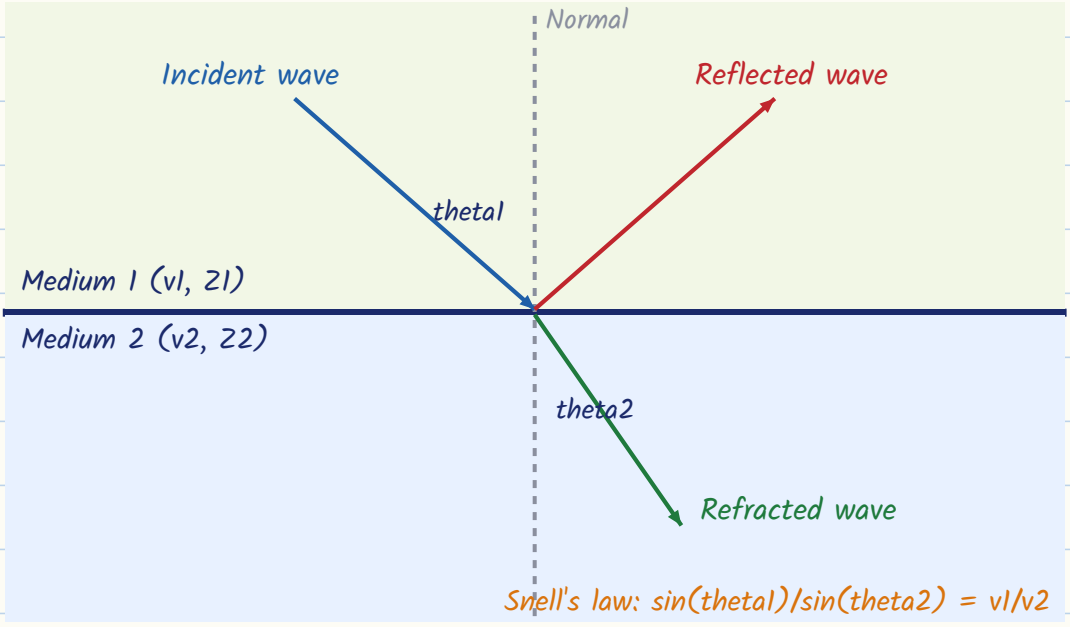


Fig: Reflection and refraction at an interface

4.3 Critical Angles

* **First critical angle:** incident angle at which the refracted longitudinal wave becomes 90 degrees (grazing) - beyond it, no longitudinal wave in the second medium.
* **Second critical angle:** incident angle at which the refracted shear wave becomes 90 degrees - beyond it, a surface wave is generated.
* Between the first and second critical angle, only a **pure shear wave** is in the test piece - the principle of the angle probe.
* For Perspex to steel: first critical angle about 27.5 deg and second about 57 deg.

4.4 Diffraction

* Bending of waves around the edge of an obstacle / small discontinuity.
* If flaw size is **much smaller than the wavelength**, the wave passes around it - not detected. So the **smallest detectable flaw is about lambda / 2**.
* Used in **TOFD (Time of Flight Diffraction)** technique - the tips of the crack emit diffracted signals which give the crack height accurately.

5. Mode Conversion

* When a wave strikes an interface **obliquely**, part of the energy changes into other wave type; for example a longitudinal wave can produce both reflected and refracted **shear** waves.
* Number and angle of waves follow Snell's law for each mode.
* Mode conversion is **useful** for making shear probes, but **harmful** when it gives false (ghost) echoes.
* In a straight beam test, longitudinal wave hitting the wall of the specimen at an angle can create shear waves and cause **spurious indications**.
* Normal incidence -> no mode conversion.

6. Attenuation

* **Attenuation** = decrease of sound intensity with distance travelled.
* Causes:
  - **Absorption** - conversion of sound energy into heat (viscous friction).

- Scattering - from grain boundaries, inclusions, porosity (strong in cast iron, austenitic steel, castings).
- Beam spreading (divergence) - geometric loss.

* Attenuation coefficient in decibel: alpha = (1/x) x 20 log10 (A1 / A2) dB per unit length.
* Intensity: I = I0 e^(-alpha x) .
* Attenuation rises with frequency and with coarse grain size - so low frequency (0.5 - 2 MHz) is used for coarse grained castings and high frequency (5 - 10 MHz) for fine grained forgings.
* Decibel (dB): gain control in the instrument; dB = 20 log10 (A2 / A1) . A change of 6 dB = double amplitude, 20 dB = ten times amplitude.

7. Sound Field of the Transducer

* Sound beam from a disc shaped crystal of diameter D has two zones:

Near Field (Fresnel zone)

* Region close to the probe with large pressure variations (maxima and minima due to interference) - flaw evaluation is unreliable here.
* Length of near field: N = D^2 / (4 lambda) = D^2 f / (4 v) .

Far Field (Fraunhofer zone)

* Beyond N, the beam diverges and intensity decreases smoothly with distance (inverse square law); testing is reliable here.
* Beam divergence (half angle theta): sin(theta) = 1.22 lambda / D - smaller crystal or lower frequency gives wider spread.
* Higher frequency or bigger crystal -> longer near field and narrower beam .

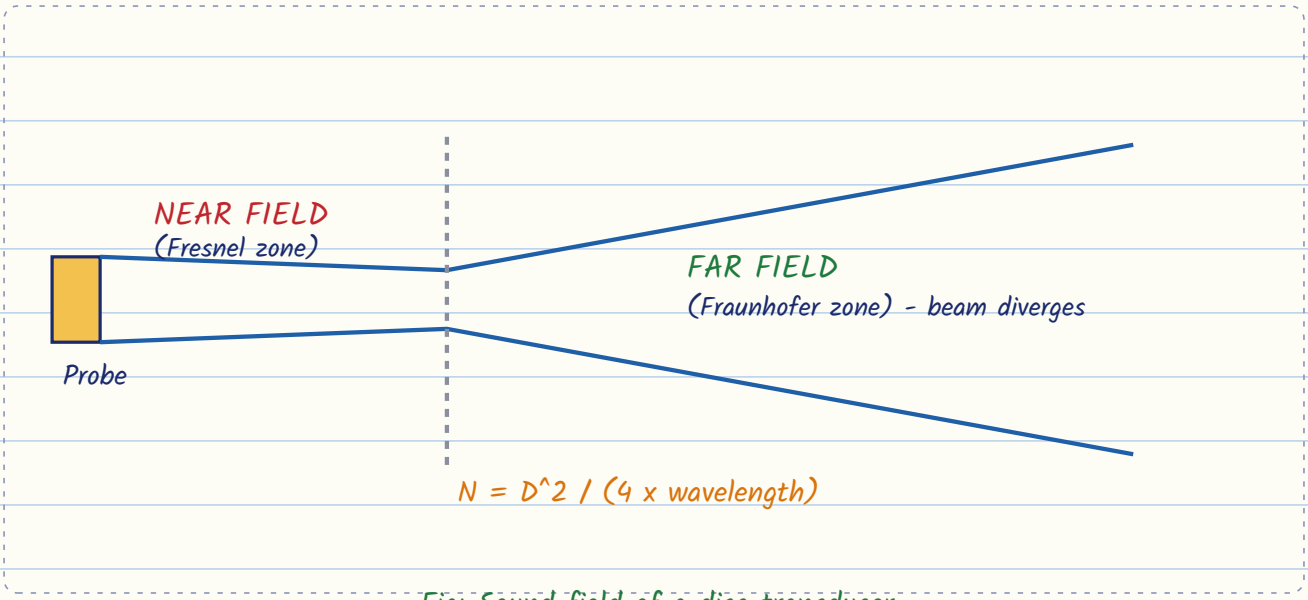


Fig: Sound field of a disc transducer

Remember:

Large diameter + high frequency = long near field, narrow beam (good directivity). Small diameter + low frequency = short near field, wide beam.

8. Piezo-electric Effect

* Certain crystals produce **electric charge when mechanically stressed** (direct effect) and **change in dimension when an electric field is applied** (inverse / converse effect).
* **Transmitter:** high voltage electrical pulse -> crystal vibrates -> ultrasound (inverse effect).
* **Receiver:** returning echo -> crystal compressed -> voltage (direct effect).
* Thickness of the crystal for resonance: $t = \lambda / 2 = v / (2f)$ - thin crystal gives high frequency.

Piezoelectric Materials

Material	Features
Quartz (natural)	Stable, hard, low efficiency; used for high frequency
Lithium sulphate	Good receiver, easily damped, water soluble
Barium titanate	High efficiency transmitter (ceramic)
Lead zirconate titanate (PZT)	Very efficient, most common, polarised ceramic
Lead metaniobate	Heavily damped, broad band, good resolution

PVDF (polymer film)	Wide bandwidth, flexible, used for special probes

* Curie temperature: above it the crystal loses piezo-electric property.

9. Ultrasonic Transducers (Probes) and Characteristics

9.1 Construction of a Probe

* **Piezoelectric element (crystal)** - the active part.
* **Backing (damping) block** - tungsten-epoxy; absorbs backward waves and **shortens the pulse** for good resolution.
* **Wear plate / protective face** - protects the crystal; matching layer improves transmission.
* **Housing and connector (BNC / Lemo)** with electrical matching network (tuning coil).

9.2 Types of Probes

* **Normal (straight beam) probe:** longitudinal wave, perpendicular to the surface. Detects laminations, forgings, thickness.
* **Angle beam probe:** crystal on a wedge; 35, 45, 60, 70 degrees; shear wave. Used for welds.
* **Dual crystal (TR) probe:** separate transmitter and receiver crystals, each tilted slightly - good for **thin sections and near surface flaws, corrosion thickness gauging** .
* **Immersion probe:** used under water with automatic scanning; may have focusing lens.
* **Focused probe:** lens gives narrow beam at a focal point - high sensitivity for small flaws.
* **Delay line probe:** plastic delay between crystal and part - for thin materials.
* **Phased array probe:** many small elements fired by software with delays to steer and focus the beam.
* **EMAT:** electro-magnetic acoustic transducer - needs no couplant; works on hot or rough surface.

9.3 Probe Characteristics

* **Sensitivity:** ability to detect small flaws (smallest echo above noise).
* **Resolution:** ability to separate two close echoes - short pulse, high frequency, good damping. Near surface resolution = **dead zone** reduction.
* **Frequency:** nominal frequency in MHz; bandwidth = (f upper - f lower) / f centre.
* **Beam spread, near field length and focal length.**
* **Damping / Q factor:** highly damped = broad band, good resolution, lower sensitivity.
* **Dead zone:** area below the surface where flaw cannot be detected due to the ringing of transmitter pulse.

Frequency	Result
Low (0.5 - 2 MHz)	More penetration, less attenuation, poor resolution - castings
Medium (2 - 5 MHz)	General purpose - welds, forgings
High (5 - 25 MHz)	Less penetration, better sensitivity - thin, fine grained

10. Ultrasonic Equipment

10.1 Flaw Detector Block Diagram

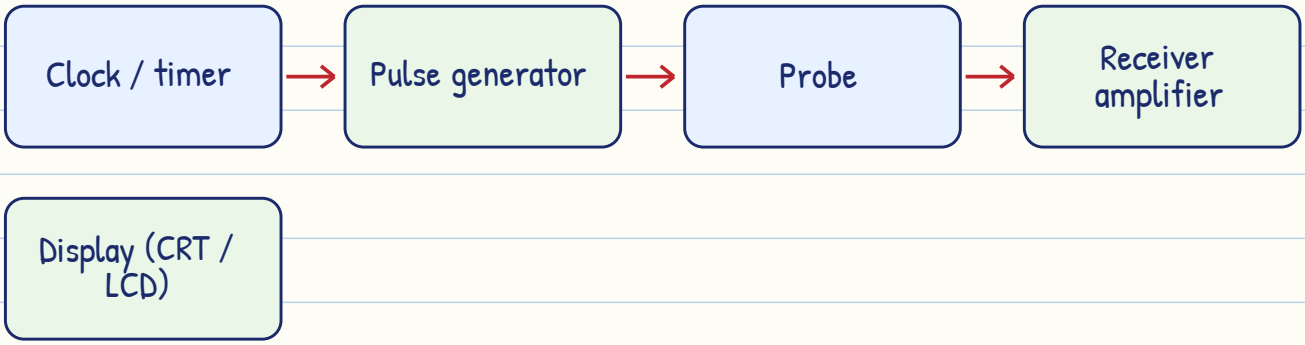


Fig: Basic block diagram of a flaw detector

* **Pulse generator:** produces short high voltage pulses (100 - 500 V) at PRF = pulse repetition frequency (50 - 2000 pulses per second).
* **Time base (sweep) generator:** moves the beam across the screen - horizontal axis = distance.
* **Receiver / amplifier:** amplifies weak echo, with gain control in dB, reject (suppression), and DAC (distance amplitude correction).
* **Display:** vertical axis = echo amplitude.

* **Gate and alarm:** monitoring section for automatic detection.

10.2 Display Types

* **A-scan:** amplitude vs time (distance) - most common; gives flaw depth and relative size.
* **B-scan:** cross-sectional view (depth vs probe position) - side view of the flaw.
* **C-scan:** plan view of the part (like X-ray) - amplitude / depth colour map by scanning x-y; used in composites.
* **D-scan, S-scan (sector scan in phased array)** .

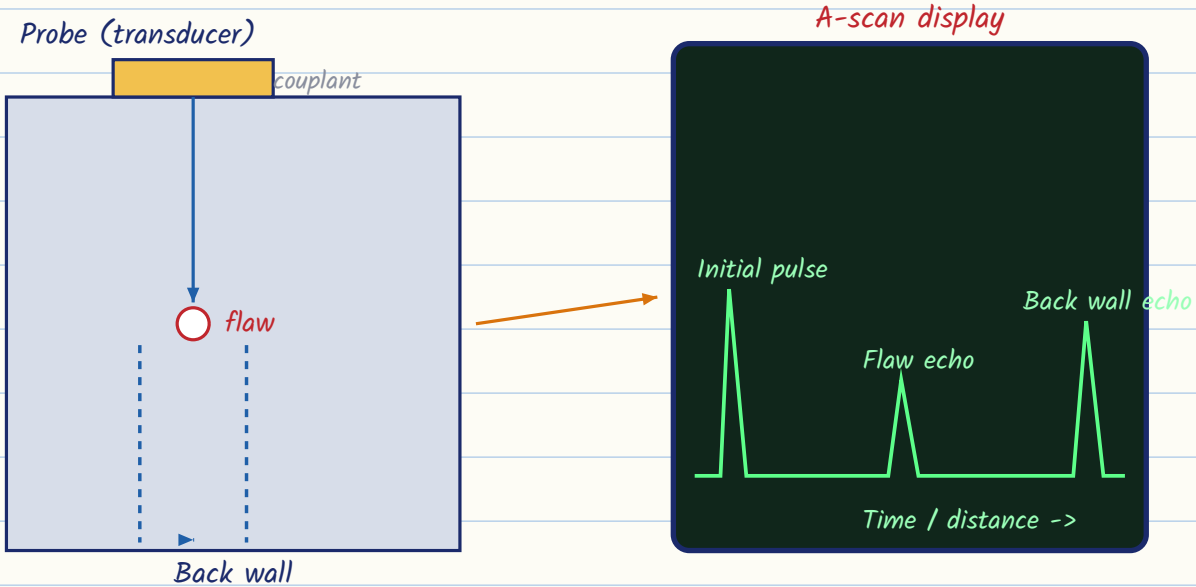


Fig: Pulse-echo method and A-scan display

10.3 Testing Techniques (Methods)

* **Pulse-echo (reflection) method:** one probe sends and receives. Most used.
* **Through transmission:** two probes on opposite sides; a flaw reduces the received signal (shadow). Needs access to both sides; no dead zone; used in automated testing.
* **Pitch-catch (two probe) technique:** separate sending and receiving probes on the same side.
* **Resonance method:** continuous wave frequency varied to find resonance -> thickness measurement.
* **Contact testing:** probe in direct contact with a thin couplant film.

* **Immersion testing:** part in a water tank; automatic scanning, better coupling, C-scan; interface echo present.
* **Gap (squirter) testing** using a water jet.

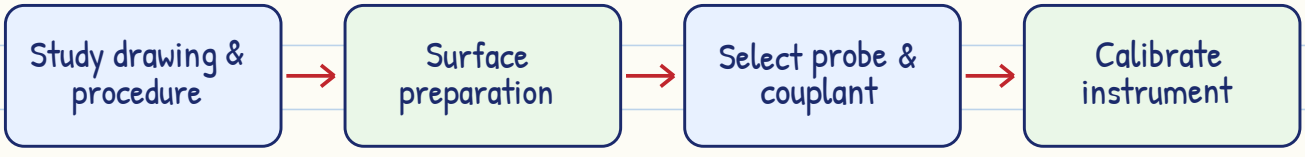
10.4 Calibration Blocks

* **V1 (IIW) block** – calibration of time base, probe index and beam angle, sensitivity.
* **V2 block** – miniature field block for angle probes.
* **Step wedge blocks** – thickness; **DAC reference blocks** with side drilled holes; ASTM basic block, area-amplitude and distance-amplitude blocks.
* Calibration is done before every test and re-checked periodically.

11. Variables Affecting Ultrasonic Test

* **Frequency and wavelength** – control sensitivity, penetration, resolution.
* **Probe diameter and type, probe angle** – control beam spread and near field.
* **Material properties:** velocity, grain size, attenuation, density, anisotropy (weld metal).
* **Surface condition:** rough surfaces, scale, paint reduce coupling (loss of echo). Use grinding, couplant of high viscosity.
* **Couplant** thickness and type.
* **Flaw characteristics:** size, shape, orientation (the best echo when flaw is **perpendicular** to beam), depth, roughness, acoustic impedance of the flaw.
* **Test piece geometry:** curvature, thin sections, edges (mode conversion, side-wall echoes).
* **Instrument settings:** gain, reject, PRF, range, pulse energy, damping, filters.
* **Temperature** – velocity changes with temperature; hot parts need special probes.
* **Operator skill** and procedure.

12. Ultrasonic Testing Procedure



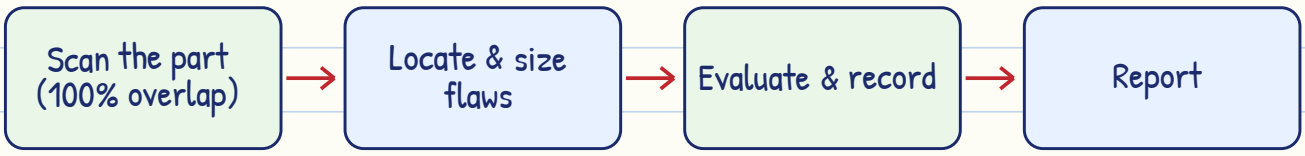


Fig: Steps in UT

* **Weld testing:** angle probes of 45, 60, 70 degrees scan the weld from both sides with zig-zag movement, using skip distance and half skip distance. Position of flaw: depth = s cos(theta), surface distance = s sin(theta), where s = sound path length.
* **Plate / forging testing:** normal probe, grid scanning to find laminations and inclusions.
* **Thickness measurement:** dual probe or digital thickness gauge (corrosion monitoring in pipes, tanks).
* **Flaw sizing techniques:** 6 dB drop (half amplitude) method for large flaws, 20 dB drop method, DGS / AVG diagram, DAC curve, TOFD.

13. Interpretation and Guidelines for Acceptance / Rejection

* Indication evaluated by **amplitude** (compared with a reference level such as DAC or a reference reflector), **length** (probe movement between the drop points), **depth and location** , and **type (planar or volumetric)** from the echo behaviour.
* Dynamic echo pattern: a **smooth, sharp, steady echo** that rises and falls rapidly with probe movement = planar flaw such as crack; a **wider echo that moves in many peaks** = porosity or slag.
* A crack gives a strong echo from one direction only; porosity gives a small echo from many directions.
* Acceptance as per codes: ASME Sec. V and VIII, ASME B31.3, AWS D1.1, API 1104, IS 4260 / IS 7666, ISO 11666.
* **Typical acceptance criteria:** all indications showing **cracks, lack of fusion or incomplete penetration are rejected regardless of size** ; others rejected when amplitude exceeds the reference level (DAC) and length exceeds the permitted value (for example one-third of weld thickness for a thin weld, up to a limit).
* Records: report with probe details, calibration block, gain, sketches, flaw position, results and name of qualified level II / III person.

Indication	Probable cause
Strong, sharp echo, moves with probe	Crack, lack of fusion
Cluster of small echoes, low amplitude	Porosity
Broad irregular echo	Slag inclusion
Echo at plate mid thickness	Lamination
Loss of back wall echo	Large defect or bad coupling / grain

14. Effectiveness and Limitations of Ultrasonic Testing

Advantages

* Very **sensitive** to planar defects such as cracks; high penetration power (several metres of steel).
* **Accurate** location and size of flaw (depth, length).
* Access to **only one surface** needed.
* **No radiation hazard** , portable equipment, instant results.
* Suitable for metals, plastics, composites, ceramics; can measure thickness and material properties (grain size, modulus).
* Automation and permanent records possible (C-scan, phased array).

Limitations

* **Skill and training** of operator is very important; interpretation is difficult.
* Surface must be **smooth and clean** ; couplant needed.
* Difficult on **rough, irregular, thin or small parts** , coarse-grained materials (cast iron, austenitic weld).
* Flaws parallel to the beam are not detected; **dead zone** near the surface.
* Needs **reference standards (calibration blocks)** .
* No permanent record in conventional A-scan.

Summary:

UT = high frequency sound, pulse-echo, uses piezo-electric probes; velocity v = f lambda; best for planar flaws; depends on couplant, impedance mismatch, attenuation and operator skill.

Important Questions

* Explain piezoelectric effect and construction of an ultrasonic probe.
* Define near field and far field. Derive N = D^2 / 4 lambda.
* Explain reflection, refraction, mode conversion and attenuation.
* Differentiate A, B and C scans. Explain pulse-echo and through-transmission methods.
* Write advantages and limitations of UT.